

2026年04月CSCA官方数学试题

CSCA Official Mathematics Examination Paper, April 2026

题目(Questions)、答案(Answers)与解析(Explanations)

Instructions: This is a multiple-choice test. For each question, choose the **only one correct answer**.

(说明：本试卷为单项选择题，每题仅有一个正确答案。)

- Let $A = 1, 3, 4, 5, 7$, $B = 1, 3, 5$ be two sets. Which of the following statements is correct?
设 $A = 1, 3, 4, 5, 7$, $B = 1, 3, 5$ 为两个集合，下列说法正确的是 ()
(A) $B \subseteq A$
(B) $B \in A$
(C) $A \in B$
(D) $A \subseteq B$
- Let $A = x \mid 3 < x < 5$, $B = x \mid x > 4$ be two sets. Then $A \cap B = ()$
设 $A = x \mid 3 < x < 5$, $B = x \mid x > 4$ 为两个集合，则 $A \cap B = ()$
(A) $x \mid 3 < x < 4$
(B) $x \mid 3 < x \leq 4$
(C) $x \mid 4 < x < 5$
(D) $x \mid 4 \leq x < 5$
- Which of the following points is in the third quadrant?
下列哪个点在第三象限? ()
(A) $(1, 2)$
(B) $(-1, 2)$
(C) $(1, -2)$
(D) $(-1, -2)$
- The solution set of the inequality $x^2 - 2x - 3 \geq 0$ is ()
不等式 $x^2 - 2x - 3 \geq 0$ 的解集是 ()
(A) $(-\infty, -1) \cup (3, +\infty)$
(B) $[-1, 3)$
(C) $(-1, 3)$
(D) $(-\infty, -1] \cup [3, +\infty)$
- The domain of the function $f(x) = \sqrt[3]{x} + 1$ is ()
函数 $f(x) = \sqrt[3]{x} + 1$ 的定义域是 ()
(A) $[1, +\infty)$
(B) $[0, +\infty)$
(C) $(-\infty, 0]$
(D) $(-\infty, +\infty)$
- In the Cartesian coordinate system, the horizontal coordinate of point M is -4 , and its distance to the x -axis is 5. The coordinates of M are ()
在平面直角坐标系中，点 M 的横坐标为 -4 ，到 x 轴的距离为 5，则点 M 的坐标是 ()
(A) $(-4, 5)$ or $(-4, -5)$
(B) $(5, -4)$

- (C) $(-5, -4)$
 (D) $(-4, 5)$ or $(-4, 3)$
7. $\sin 120^\circ = ()$
 $\sin 120^\circ = ()$
 (A) $-\frac{1}{2}$
 (B) $\frac{1}{2}$
 (C) $-\frac{\sqrt{3}}{2}$
 (D) $\frac{\sqrt{3}}{2}$
8. Which of the following functions is an odd function?
 下列哪个函数是奇函数? ()
 (A) $y = x^3 + 1$
 (B) $y = \sqrt[3]{x} + 2$
 (C) $y = x^2 + 3$
 (D) $y = -x$
9. The slope of the straight line passing through points $A(-1, 1)$ and $B(2, 3)$ is ()
 过点 $A(-1, 1)$ 、 $B(2, 3)$ 的直线的斜率是 ()
 (A) $-\frac{2}{3}$
 (B) $\frac{2}{3}$
 (C) $-\frac{3}{2}$
 (D) $\frac{3}{2}$
10. The inverse function of the function $y = 10x + 3$ is ()
 函数 $y = 10x + 3$ 的反函数是 ()
 (A) $y = \frac{x-10}{3}$
 (B) $y = \frac{x+10}{3}$
 (C) $y = \frac{x-3}{10}$
 (D) $y = \frac{x+3}{10}$
11. Which of the following functions is a decreasing function on $(-\infty, +\infty)$?
 下列哪个函数在 $(-\infty, +\infty)$ 上是减函数? ()
 (A) $y = x^3$
 (B) $y = -\frac{1}{x}$
 (C) $y = -x + 5$
 (D) $y = x^2 + 1$
12. The solution set of the inequality $\frac{2x+1}{x-1} < 0$ is ()
 不等式 $\frac{2x+1}{x-1} < 0$ 的解集是 ()
 (A) $(-\frac{1}{2}, 1]$
 (B) $(-\infty, -\frac{1}{2}] \cup [1, +\infty)$
 (C) $(-\infty, -\frac{1}{2}) \cup (1, +\infty)$
 (D) $(-\frac{1}{2}, 1)$
13. If $-1, a, 15$ form an arithmetic sequence, then $a = ()$
 若 $-1, a, 15$ 构成等差数列, 则 $a = ()$
 (A) 11
 (B) 7
 (C) 5
 (D) 9
14. If the angle of inclination of a straight line l is 60° , then the slope of l is ()
 若直线 l 的倾斜角为 60° , 则直线 l 的斜率为 ()
 (A) $\frac{\sqrt{3}}{3}$
 (B) $\sqrt{3}$

- (C) $-\frac{1}{2}$
 (D) $\frac{1}{2}$
15. If a_n is an arithmetic sequence, $a_1 = 1$, $a_2 = 3$, then $a_5 = ()$
 若数列 a_n 为等差数列, 且 $a_1 = 1$, $a_2 = 3$, 则 $a_5 = ()$
 (A) 11
 (B) 9
 (C) 5
 (D) 7
16. If point $P(x, y)$ is on the terminal side of angle α , and the distance from P to the origin is $r(r \neq 0)$, then $\cos \alpha = ()$
 若点 $P(x, y)$ 在角 α 的终边上, 且点 P 到原点的距离为 $r(r \neq 0)$, 则 $\cos \alpha = ()$
 (A) $\frac{x}{y}$
 (B) $\frac{y}{x}$
 (C) $\frac{y}{r}$
 (D) $\frac{x}{r}$
17. If the general term of sequence a_n is $a_n = 2n - 1$, which of the following statements is correct?
 若数列 a_n 的通项公式为 $a_n = 2n - 1$, 下列说法正确的是()
 (A) a_n is an arithmetic sequence
 (B) a_n is a geometric sequence
 (C) a_n is neither an arithmetic nor a geometric sequence
 (D) a_n is both an arithmetic and a geometric sequence
18. Suppose the distance from point $A(0, 3)$ to point $B(x, -1)$ is 5. Then $x = ()$
 若点 $A(0, 3)$ 到点 $B(x, -1)$ 的距离为5, 则 $x = ()$
 (A) 3
 (B) 2
 (C) -3
 (D) ± 3
19. If a straight line l has a slope of -3 and passes through the point $(1, 2)$, then the equation of l is()
 若直线 l 的斜率为 -3 , 且过点 $(1, 2)$, 则直线 l 的方程是()
 (A) $x + 3y + 5 = 0$
 (B) $3x + y + 1 = 0$
 (C) $3x + y - 5 = 0$
 (D) $x + 3y - 7 = 0$
20. Suppose a circle has a radius of 3 and its center is at $(-1, 2)$. Then the equation of the circle is()
 若一个圆的半径为3, 圆心坐标为 $(-1, 2)$, 则该圆的方程是()
 (A) $x^2 + y^2 + 2x - 4y - 4 = 0$
 (B) $x^2 + y^2 - 2x - 4y - 4 = 0$
 (C) $x^2 + y^2 + 2x + 4y - 4 = 0$
 (D) $x^2 + y^2 + 2x - 4y + 4 = 0$
21. If $\cos \alpha = \frac{2}{3}$ and $0 < \alpha < \frac{\pi}{2}$, then $\tan \alpha = ()$
 若 $\cos \alpha = \frac{2}{3}$, 且 $0 < \alpha < \frac{\pi}{2}$, 则 $\tan \alpha = ()$
 (A) $\frac{1}{2}$
 (B) $\frac{\sqrt{5}}{2}$
 (C) $\frac{2}{3}$
 (D) $\frac{\sqrt{3}}{2}$

22. If a_n is a geometric sequence, $a_1 = -48$, $a_4 = 6$, then the common ratio $q = ()$
 若数列 a_n 为等比数列, 且 $a_1 = -48$, $a_4 = 6$, 则公比 $q = ()$
 (A) $\frac{1}{2}$
 (B) -2
 (C) 2
 (D) $-\frac{1}{2}$
23. If a circle passes through point $A(2, 2)$ and its center is at $(2, 0)$, then the equation of the circle is $()$
 若一个圆过点 $A(2, 2)$, 且圆心为 $(2, 0)$, 则该圆的方程是 $()$
 (A) $(x - 2)^2 + y^2 = 2$
 (B) $(x - 2)^2 + (y - 2)^2 = 4$
 (C) $x^2 + (y - 2)^2 = 2$
 (D) $(x - 2)^2 + y^2 = 4$
24. If a, b, c are rational numbers and $a > b$, which of the following inequalities is correct?
 若 a, b, c 为有理数, 且 $a > b$, 则下列不等式正确的是 $()$
 (A) $a - c < b - c$
 (B) $ac > bc$
 (C) $\frac{a}{c} > \frac{b}{c}$
 (D) $a + c > b + c$
25. If the exponential function $y = a^x$ is monotonic decreasing, then the range of a is $()$
 若指数函数 $y = a^x$ 单调递减, 则 a 的取值范围是 $()$
 (A) $a > 1$
 (B) $a < 1$
 (C) $a > 0$
 (D) $0 < a < 1$
26. If $\sin \alpha = \frac{3}{4}$, then $\cos 2\alpha = ()$
 若 $\sin \alpha = \frac{3}{4}$, 则 $\cos 2\alpha = ()$
 (A) $-\frac{1}{8}$
 (B) $\frac{1}{8}$
 (C) $-\frac{7}{16}$
 (D) $\frac{7}{16}$
27. The coordinates of the intersection of lines $l_1 : 3x - y + 2 = 0$ and $l_2 : 4x - y + 3 = 0$ are $()$
 直线 $l_1 : 3x - y + 2 = 0$ 与直线 $l_2 : 4x - y + 3 = 0$ 的交点坐标是 $()$
 (A) $(1, 1)$
 (B) $(-1, 1)$
 (C) $(1, -1)$
 (D) $(-1, -1)$
28. Given the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{3} = 1 (a > 0)$ with eccentricity $e = 2$, then $a = ()$
 已知双曲线 $\frac{x^2}{a^2} - \frac{y^2}{3} = 1 (a > 0)$ 的离心率 $e = 2$, 则 $a = ()$
 (A) 1
 (B) 1
 (C) 2
 (D) $\frac{\sqrt{6}}{2}$
29. Given the sequence $a_n : \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, its general term $a_n = ()$
 已知数列 $a_n : \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, 其通项公式 $a_n = ()$
 (A) $(-1)^{n-1} \frac{1}{2^n}$
 (B) $(-1)^n \frac{1}{2^n}$

- (C) $\frac{1}{2^{n-1}}$
 (D) $\frac{1}{2^n}$
30. $(\log_3 27)(\log_4 16) = ()$
 $(\log_3 27)(\log_4 16) = ()$
 (A) 5
 (B) $\frac{1}{6}$
 (C) 6
 (D) $\frac{2}{3}$
31. The minimum value of the function $y = \cos 5x + 4$ is ()
 函数 $y = \cos 5x + 4$ 的最小值是 ()
 (A) 4
 (B) 5
 (C) 2
 (D) 3
32. If the parabola $y = ax^2$ passes through the point $(1, 2)$, then $a = ()$
 若抛物线 $y = ax^2$ 经过点 $(1, 2)$, 则 $a = ()$
 (A) 1
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) 2
33. Which of the following straight lines is perpendicular to the line $-2x + y - 2 = 0$?
 下列哪条直线与直线 $-2x + y - 2 = 0$ 垂直? ()
 (A) $x + y - 2 = 0$
 (B) $2x + y - 2 = 0$
 (C) $-x + 2y - 2 = 0$
 (D) $x + 2y - 2 = 0$
34. Given $\cos \alpha = -\frac{1}{2}$, $\alpha \in (\frac{\pi}{2}, \pi)$, then $\sin \frac{\alpha}{2} = ()$
 已知 $\cos \alpha = -\frac{1}{2}$, $\alpha \in (\frac{\pi}{2}, \pi)$, 则 $\sin \frac{\alpha}{2} = ()$
 (A) $\frac{\sqrt{3}}{2}$
 (B) $-\frac{\sqrt{3}}{2}$
 (C) $-\frac{1}{2}$
 (D) $\frac{1}{2}$
35. $\cos 75^\circ = ()$
 $\cos 75^\circ = ()$
 (A) $\frac{\sqrt{6}-\sqrt{2}}{4}$
 (B) $-\frac{\sqrt{6}+\sqrt{2}}{4}$
 (C) $\frac{\sqrt{2}-\sqrt{6}}{4}$
 (D) $\frac{\sqrt{6}+\sqrt{2}}{4}$
36. If the domain of the function $y = f(x)$ is $(-1, 0)$, then the domain of the function $y = f(x^2 - 1)$ is ()
 若函数 $y = f(x)$ 的定义域为 $(-1, 0)$, 则函数 $y = f(x^2 - 1)$ 的定义域是 ()
 (A) $(-1, 0) \cup (0, 1)$
 (B) $(-1, 0)$
 (C) $(-1, 1)$
 (D) $[-1, 1]$
37. Given $\cos \alpha = -\frac{7}{25}$ and $\frac{\pi}{2} < \alpha < \pi$, then $\cos \frac{\alpha}{2} = ()$
 已知 $\cos \alpha = -\frac{7}{25}$, 且 $\frac{\pi}{2} < \alpha < \pi$, 则 $\cos \frac{\alpha}{2} = ()$
 (A) $\frac{3}{5}$
 (B) $-\frac{3}{5}$

(C) $-\frac{4}{5}$

(D) $\frac{4}{5}$

38. Which of the following identities is correct?

下列等式正确的是 ()

(A) $\cos(\pi + \alpha) = \cos \alpha$

(B) $\sin(\pi + \alpha) = \sin \alpha$

(C) $\cos\left(\frac{\pi}{2} + \alpha\right) = \sin \alpha$

(D) $\sin\left(\frac{\pi}{2} + \alpha\right) = \cos \alpha$

39. The minimum positive period of the function $y = 2 \tan\left(3x + \frac{\pi}{4}\right)$ is ()

函数 $y = 2 \tan\left(3x + \frac{\pi}{4}\right)$ 的最小正周期是 ()

(A) 2π

(B) $\frac{\pi}{4}$

(C) π

(D) $\frac{\pi}{3}$

40. Let ellipse C be centered at the origin with its foci on the y -axis, focal length 4, and eccentricity $\frac{2}{3}$. Then the equation of C is ()

椭圆 C 以原点为中心, 焦点在 y 轴上, 焦距为 4, 离心率为 $\frac{2}{3}$, 则 C 的方程是 ()

(A) $\frac{x^2}{12} + \frac{y^2}{8} = 1$

(B) $\frac{x^2}{5} + \frac{y^2}{9} = 1$

(C) $\frac{x^2}{9} + \frac{y^2}{5} = 1$

(D) $\frac{x^2}{8} + \frac{y^2}{12} = 1$

41. If sequence a_n satisfies $a_n = (-1)^n 2a_{n-1} (n \geq 2)$ and $a_1 = \frac{1}{3}$, then $a_4 =$ ()

若数列 a_n 满足 $a_n = (-1)^n 2a_{n-1} (n \geq 2)$, 且 $a_1 = \frac{1}{3}$, 则 $a_4 =$ ()

(A) $\frac{8}{3}$

(B) $-\frac{8}{3}$

(C) $-\frac{16}{3}$

(D) $\frac{16}{3}$

42. The domain of the function $y = \frac{1}{\lg|x-5|}$ is ()

函数 $y = \frac{1}{\lg|x-5|}$ 的定义域是 ()

(A) $(-\infty, 5) \cup (5, +\infty)$

(B) $(-\infty, 4) \cup (4, 5) \cup (5, 6) \cup (6, +\infty)$

(C) $(-\infty, 5) \cup (5, 6) \cup (6, +\infty)$

(D) $(-\infty, 4) \cup (4, +\infty)$

43. Given ellipse C has the same foci as the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$, and the length of its minor axis is $4\sqrt{5}$. Then the equation of ellipse C is ()

已知椭圆 C 与椭圆 $\frac{x^2}{4} + \frac{y^2}{9} = 1$ 有相同的焦点, 且 C 的短轴长为 $4\sqrt{5}$, 则椭圆 C 的方程是 ()

(A) $\frac{x^2}{25} + \frac{y^2}{20} = 1$

(B) $\frac{x^2}{20} + \frac{y^2}{45} = 1$

(C) $\frac{x^2}{80} + \frac{y^2}{85} = 1$

(D) $\frac{x^2}{20} + \frac{y^2}{25} = 1$

44. Let $ABCD$ be a square, and point E is the midpoint of CD . Given $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{AD} = \vec{b}$,

then vector $\overrightarrow{BE} =$ ()

设正方形 $ABCD$ 中, 点 E 为 CD 边的中点。已知 $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AD} = \vec{b}$, 则向量 $\overrightarrow{BE} =$ ()

(A) $\vec{b} + \frac{1}{2}\vec{a}$

(B) $\vec{a} - \frac{1}{2}\vec{b}$

(C) $\vec{b} - \frac{1}{2}\vec{a}$

(D) $\vec{a} + \frac{1}{2}\vec{b}$

45. Let the sum of the first n terms of a geometric sequence a_n be S_n , and $2S_n = 3a_{n+1} - 3$ ($n = 1, 2, \dots$). Then $S_n =$ ()
 设等比数列 a_n 的前 n 项和为 S_n , 且 $2S_n = 3a_{n+1} - 3$ ($n = 1, 2, \dots$), 则 $S_n =$ ()
 (A) $\frac{2}{3} \left(\frac{5}{3}\right)^n - \frac{2}{3}$
 (B) $\frac{2}{3} \left(\frac{3}{5}\right)^n - \frac{2}{3}$
 (C) $\frac{3}{2} \left(\frac{3}{5}\right)^n - \frac{3}{2}$
 (D) $\frac{3}{2} \left(\frac{5}{3}\right)^n - \frac{3}{2}$
46. If a complex number z satisfies the equation $(z - 2i)(2 - i) = 5$, then $z =$ ()
 若复数 z 满足方程 $(z - 2i)(2 - i) = 5$, 则 $z =$ ()
 (A) $3 + 2i$
 (B) $2 + 3i$
 (C) $3 - 2i$
 (D) $2 - 3i$
47. If the sum of the first n terms of sequence a_n is $S_n = \frac{2}{3}a_n + \frac{1}{3}$, then the general term $a_n =$ ()
 若数列 a_n 的前 n 项和 $S_n = \frac{2}{3}a_n + \frac{1}{3}$, 则该数列的通项公式 $a_n =$ ()
 (A) $(-2)^n$
 (B) $\left(-\frac{1}{2}\right)^{n-1}$
 (C) $\left(\frac{1}{2}\right)^{n-1}$
 (D) $(-2)^{n-1}$
48. There are 12 identical balls in an opaque bag: red, black, yellow and green balls. Xiaohong draws a ball at random. The probability of drawing a red ball is $\frac{1}{3}$, the probability of drawing a yellow or green ball is $\frac{5}{12}$, and the probability of drawing a black or yellow ball is also $\frac{5}{12}$. The probabilities of drawing a yellow ball, a black ball and a green ball are respectively ()
 一个不透明袋子中有 12 个形状相同的球: 红球、黑球、黄球、绿球。小红随机摸出一个球, 摸到红球的概率为 $\frac{1}{3}$, 摸到黄球或绿球的概率为 $\frac{5}{12}$, 摸到黑球或黄球的概率也为 $\frac{5}{12}$ 。则摸到黄球、黑球、绿球的概率分别是 ()
 (A) $\frac{1}{3}, \frac{1}{4}, \frac{1}{6}$
 (B) $\frac{1}{5}, \frac{1}{4}, \frac{1}{6}$
 (C) $\frac{1}{6}, \frac{1}{4}, \frac{1}{4}$
 (D) $\frac{1}{4}, \frac{1}{5}, \frac{1}{6}$

Answer标准答案速查表

1.A 2.C 3.D 4.D 5.D 6.A 7.D 8.D 9.B 10.C
 11.C 12.D 13.B 14.B 15.B 16.D 17.A 18.D 19.C 20.A
 21.B 22.D 23.D 24.D 25.D 26.A 27.D 28.A 29.A 30.C
 31.D 32.D 33.D 34.A 35.A 36.A 37.A 38.D 39.D 40.B
 41.B 42.B 43.D 44.C 45.D 46.B 47.D 48.C

1. Answer: A

- English Explanation: All elements of set $B = 1, 3, 5$ are in set $A = 1, 3, 4, 5, 7$, so $B \subseteq A$ (subset definition).
- 中文解析: 集合 B 的所有元素都属于 A , 根据子集定义, $B \subseteq A$ 成立。

2. Answer: C

- English Explanation: $A \cap B$ takes the common part of $3 < x < 5$ and $x > 4$, so $4 < x < 5$.
- 中文解析: 集合交集取同时满足 $3 < x < 5$ 和 $x > 4$ 的部分, 结果为 $4 < x < 5$ 。

3. Answer: D

- English Explanation: The third quadrant requires $x < 0$ and $y < 0$, only $(-1, -2)$ meets this.
- 中文解析: 第三象限点的横、纵坐标均为负数, 仅 $(-1, -2)$ 符合。

4. Answer: D

- English Explanation: Factor: $x^2 - 2x - 3 = (x - 3)(x + 1) \geq 0$, solution: $x \leq -1$ or $x \geq 3$.
- 中文解析: 因式分解得 $(x - 3)(x + 1) \geq 0$, 解集为 $(-\infty, -1] \cup [3, +\infty)$ 。

5. Answer: D

- English Explanation: The cube root function $\sqrt[3]{x}$ is defined for all real numbers.
- 中文解析: 立方根函数对全体实数有定义, 定义域为 $(-\infty, +\infty)$ 。

6. Answer: A

- English Explanation: The x -coordinate is -4 , distance to x -axis is $|y| = 5$, so $y = \pm 5$.
- 中文解析: 横坐标为 -4 , 到 x 轴距离为纵坐标绝对值, $y = \pm 5$, 坐标为 $(-4, 5)$ 或 $(-4, -5)$ 。

7. Answer: D

- English Explanation: $\sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$.
- 中文解析: $\sin 120^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}$ 。

8. Answer: D

- English Explanation: Odd function: $f(-x) = -f(x)$. For $y = -x$, $f(-x) = x = -f(x)$, so it is odd.
- 中文解析: 奇函数满足 $f(-x) = -f(x)$, 仅 $y = -x$ 符合该定义。

9. Answer: B

- English Explanation: Slope formula: $k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{2 - (-1)} = \frac{2}{3}$.
- 中文解析: 斜率公式 $k = \frac{3 - 1}{2 + 1} = \frac{2}{3}$ 。

10. Answer: C

- English Explanation: Swap x and y : $x = 10y + 3$, solve for y : $y = \frac{x - 3}{10}$.
- 中文解析: 反函数求解: 互换 x 、 y 得 $x = 10y + 3$, 解得 $y = \frac{x - 3}{10}$ 。

11. Answer: C

- English Explanation: $y = -x + 5$ is a linear function with slope $-1 < 0$, strictly decreasing on $\mathbb{R}$.
- 中文解析: $y = -x + 5$ 是斜率为负的一次函数, 在 $\mathbb{R}$ 上单调递减。

12. Answer: D

- English Explanation: $\frac{2x+1}{x-1} < 0$ means numerator and denominator have opposite signs, so $-\frac{1}{2} < x < 1$.
- 中文解析: 分式不等式分子分母异号时成立, 解得 $-\frac{1}{2} < x < 1$ 。

13. Answer: B

- English Explanation: Arithmetic sequence: $2a = -1 + 15 \Rightarrow a = 7$.
- 中文解析: 等差中项性质: $2a = -1 + 15$, 得 $a = 7$ 。

14. Answer: B

- English Explanation: Slope $k = \tan \theta$, $\theta = 60^\circ$, so $k = \tan 60^\circ = \sqrt{3}$.
- 中文解析: 直线斜率 $k = \tan$ 倾斜角, $\tan 60^\circ = \sqrt{3}$ 。

15. Answer: B

- English Explanation: Common difference $d = 2$, $a_5 = a_1 + 4d = 1 + 8 = 9$.
- 中文解析: 公差 $d = 2$, $a_5 = 1 + 4 \times 2 = 9$ 。

16. Answer: D

- English Explanation: By definition: $\cos \alpha = \frac{x}{r}$ ($r = \sqrt{x^2 + y^2}$).
- 中文解析: 余弦定义: $\cos \alpha = \frac{\text{横坐标}}{\text{点到原点距离}} = \frac{x}{r}$ 。

17. Answer: A

- English Explanation: $a_{n+1} - a_n = 2$ (constant), so a_n is an arithmetic sequence.
- 中文解析: 后项减前项为常数2, 该数列为等差数列。

18. Answer: D

- English Explanation: Distance formula: $\sqrt{x^2 + (-1-3)^2} = 5 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$.
- 中文解析: 两点间距离公式 $\sqrt{x^2 + (-4)^2} = 5$, 解得 $x = \pm 3$ 。

19. Answer: C

- English Explanation: Point-slope form: $y - 2 = -3(x - 1)$, simplify to $3x + y - 5 = 0$.
- 中文解析: 点斜式 $y - 2 = -3(x - 1)$, 整理得 $3x + y - 5 = 0$ 。

20. Answer: A

- English Explanation: Standard circle equation: $(x + 1)^2 + (y - 2)^2 = 9$, expand to $x^2 + y^2 + 2x - 4y - 4 = 0$.
- 中文解析: 圆标准方程 $(x + 1)^2 + (y - 2)^2 = 9$, 展开后为 $x^2 + y^2 + 2x - 4y - 4 = 0$ 。

21. Answer: B

- English Explanation: $\sin \alpha = \sqrt{1 - (\frac{2}{3})^2} = \frac{\sqrt{5}}{3}$, $\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\sqrt{5}}{2}$.
- 中文解析: 同角三角函数关系, $\sin \alpha = \frac{\sqrt{5}}{3}$, $\tan \alpha = \frac{\sqrt{5}}{2}$ 。

22. Answer: D

- English Explanation: $a_4 = a_1 q^3 \Rightarrow 6 = -48q^3 \Rightarrow q^3 = -\frac{1}{8} \Rightarrow q = -\frac{1}{2}$.

- 中文解析：等比数列通项 $a_4 = a_1 q^3$ ，解得公比 $q = -\frac{1}{2}$ 。

23. Answer: D

- English Explanation: Radius $r = \sqrt{(2-2)^2 + (2-0)^2} = 2$, equation: $(x-2)^2 + y^2 = 4$.
- 中文解析：半径 $r = 2$ ，圆方程为 $(x-2)^2 + y^2 = 4$ 。

24. Answer: D

- English Explanation: Inequality property: if $a > b$, then $a + c > b + c$.
- 中文解析：不等式两边加同一个数，不等号方向不变。

25. Answer: D

- English Explanation: Exponential function $y = a^x$ is decreasing iff $0 < a < 1$.
- 中文解析：指数函数单调递减的充要条件是 $0 < a < 1$ 。

26. Answer: A

- English Explanation: Double-angle formula: $\cos 2\alpha = 1 - 2\sin^2 \alpha = 1 - 2 \times \left(\frac{3}{4}\right)^2 = -\frac{1}{8}$.
- 中文解析：二倍角公式 $\cos 2\alpha = 1 - 2\sin^2 \alpha = -\frac{1}{8}$ 。

27. Answer: D

- English Explanation: Solve $\begin{cases} 3x - y = -2 \\ 4x - y = -3 \end{cases}$, get $x = -1, y = -1$, intersection $(-1, -1)$.
- 中文解析：联立直线方程，解得交点为 $(-1, -1)$ 。

28. Answer: A

- English Explanation: Hyperbola: $e = \frac{c}{a} = 2, c^2 = a^2 + 3 \Rightarrow 4a^2 = a^2 + 3 \Rightarrow a = 1$.
- 中文解析：双曲线离心率 $e = 2$ ，结合 $c^2 = a^2 + b^2$ ，解得 $a = 1$ 。

29. Answer: A

- English Explanation: The sequence is $\frac{1}{2}, -\frac{1}{4}, \frac{1}{8} \dots$, general term: $a_n = (-1)^{n-1} \frac{1}{2^n}$.
- 中文解析：数列通项公式为 $a_n = (-1)^{n-1} \frac{1}{2^n}$ 。

30. Answer: C

- English Explanation: $\log_3 27 = 3, \log_4 16 = 2$, product: $3 \times 2 = 6$.
- 中文解析： $\log_3 27 = 3, \log_4 16 = 2$ ，乘积为6。

31. Answer: D

- English Explanation: The minimum of $\cos 5x$ is -1 , so $y_{\min} = -1 + 4 = 3$.
- 中文解析： $\cos 5x$ 最小值为 -1 ，函数最小值为3。

32. Answer: D

- English Explanation: Substitute $(1, 2)$ into $y = ax^2$: $2 = a \times 1^2 \Rightarrow a = 2$.
- 中文解析：代入点 $(1, 2)$ ，解得 $a = 2$ 。

33. Answer: D

- English Explanation: Slope of $-2x + y - 2 = 0$ is 2, perpendicular slope is $-\frac{1}{2}$, matching $x + 2y - 2 = 0$.
- 中文解析: 原直线斜率为2, 垂线斜率为 $-\frac{1}{2}$, 对应直线 $x + 2y - 2 = 0$ 。

34. Answer: A

- English Explanation: $\frac{\alpha}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, $\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}} = \frac{\sqrt{3}}{2}$.
- 中文解析: 半角公式, $\sin \frac{\alpha}{2} = \frac{\sqrt{3}}{2}$ 。

35. Answer: A

- English Explanation: $\cos 75^\circ = \cos(45^\circ + 30^\circ) = \frac{\sqrt{6}-\sqrt{2}}{4}$.
- 中文解析: 和角公式, $\cos 75^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}$ 。

36. Answer: A

- English Explanation: Domain condition:
 $-1 < x^2 - 1 < 0 \Rightarrow 0 < x^2 < 1 \Rightarrow x \in (-1, 0) \cup (0, 1)$.
- 中文解析: 复合函数定义域限制, 解得 $x \in (-1, 0) \cup (0, 1)$ 。

37. Answer: A

- English Explanation: $\frac{\alpha}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$, $\cos \frac{\alpha}{2} = \sqrt{\frac{1+\cos \alpha}{2}} = \frac{3}{5}$.
- 中文解析: 半角公式, $\cos \frac{\alpha}{2} = \frac{3}{5}$ 。

38. Answer: D

- English Explanation: Induction formula: $\sin(\frac{\pi}{2} + \alpha) = \cos \alpha$ is correct.
- 中文解析: 诱导公式 $\sin(\frac{\pi}{2} + \alpha) = \cos \alpha$ 正确。

39. Answer: D

- English Explanation: Period of $\tan(\omega x + \varphi)$ is $\frac{\pi}{|\omega|}$, so $T = \frac{\pi}{3}$.
- 中文解析: 正切函数最小正周期 $T = \frac{\pi}{|\omega|} = \frac{\pi}{3}$ 。

40. Answer: B

- English Explanation: Foci on y -axis, $2c = 4 \Rightarrow c = 2$, $e = \frac{c}{b} = \frac{2}{3} \Rightarrow b = 3$, $a^2 = 5$, equation: $\frac{x^2}{5} + \frac{y^2}{9} = 1$.
- 中文解析: 焦点在 y 轴, $c = 2$, $b = 3$, 椭圆方程为 $\frac{x^2}{5} + \frac{y^2}{9} = 1$ 。

41. Answer: B

- English Explanation: $a_1 = \frac{1}{3}$, $a_2 = -\frac{2}{3}$, $a_3 = -\frac{4}{3}$, $a_4 = -\frac{8}{3}$.
- 中文解析: 递推计算: $a_1 = \frac{1}{3}$, $a_2 = -\frac{2}{3}$, $a_3 = -\frac{4}{3}$, $a_4 = -\frac{8}{3}$ 。

42. Answer: B

- English Explanation: Conditions: $|x - 5| > 0$ and $\lg |x - 5| \neq 0 \Rightarrow x \neq 4, 5, 6$.
- 中文解析: 真数大于0且对数不为0, 得 $x \neq 4, 5, 6$ 。

43. Answer: D

- English Explanation: Same foci: $c^2 = 5$, minor axis $2b = 4\sqrt{5} \Rightarrow b = 2\sqrt{5}, a^2 = 25$, equation: $\frac{x^2}{20} + \frac{y^2}{25} = 1$.
- 中文解析: 同焦点椭圆, 短轴长 $4\sqrt{5}$, 方程为 $\frac{x^2}{20} + \frac{y^2}{25} = 1$ 。

44. Answer: C

- English Explanation: $\overrightarrow{BE} = \overrightarrow{BC} + \overrightarrow{CE} = \vec{b} - \frac{1}{2}\vec{a}$.
- 中文解析: 向量分解: $\overrightarrow{BE} = \overrightarrow{AD} - \frac{1}{2}\overrightarrow{AB} = \vec{b} - \frac{1}{2}\vec{a}$ 。

45. Answer: D

- English Explanation: Derive $a_1 = 1, q = \frac{5}{3}$, sum formula: $S_n = \frac{3}{2}(\frac{5}{3})^n - \frac{3}{2}$.
- 中文解析: 等比数列前 n 项和公式, 得 $S_n = \frac{3}{2}(\frac{5}{3})^n - \frac{3}{2}$ 。

46. Answer: B

- English Explanation: $z - 2i = \frac{5}{2-i} = 2 + i \Rightarrow z = 2 + 3i$.
- 中文解析: 复数运算: $z - 2i = 2 + i$, 解得 $z = 2 + 3i$ 。

47. Answer: D

- English Explanation: $a_1 = 1, a_n = -2a_{n-1}$, general term: $a_n = (-2)^{n-1}$.
- 中文解析: 由前 n 项和推得等比数列, 通项 $a_n = (-2)^{n-1}$ 。

48. Answer: C

- English Explanation: Calculate probabilities: $P(\text{yellow}) = \frac{1}{6}, P(\text{black}) = \frac{1}{4}, P(\text{green}) = \frac{1}{4}$.
- 中文解析: 概率计算: 黄球 $\frac{1}{6}$, 黑球 $\frac{1}{4}$, 绿球 $\frac{1}{4}$ 。